

GPS F3X Tracker Pylon-racing for Ethos Version 1.2

Installation guide and user manual

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1. General Description

GPS F3X Tracker Pylon-racing for Ethos is the LUA application for FrSky and Kavan transmitters with Ethos operating system. It supports training of pylon racing events (FAI F3D, F3E, F3R, F3T, some other national classes, e.g. EF1, E2K, ...) on the standard 3 pylon courses but also on simple 2 pylon ones. Whole competition event is supported, beginning with start initiated by the Start race switch or by crossing over the start/finish line and counting and time measuring of preconfigured number of legs. Via a GPS sensor the turn lines are identified and an acoustic signal is given while crossing them. A supported GPS sensor must be placed in the model and connected and configured to telemetry subsystem.

The application is based on the GPS F3X(F3F/F3B) Tracker for Ethos. Its code is published under the MIT License.

This manual describes how to install, configure and use the GPS F3X Tracker Pylon-racing.

2. Requirements

- GPS sensor: RCGPS-F3x/RCGPS-F3x-Slim ([CASSINI TECHNOLOGY](#)), SM-Modelbau GPS-Logger 3 ([GPS-Logger 3 - SM-Modellbau | RC Telemetrie Sensoren](#)), FrSky GPS ADV and generally any other GPS with data rate 10Hz are supported. We recommend RCGPS-F3x as it was designed for F3x, fully fits to the GPS F3X Tracker and its satellite positioning and navigation information remains within an acceptable accuracy range when more than 20 satellite signals remain locked during short period of high acceleration in curves
- Ethos: versions 1.6.2 and newer are supported (previous versions have various issues in areas used by the application). Ethos versions 1.6.4 and 26.x fix some other issues, which are but not critical for operation
- Transmitter: The application supports units with touchscreen resolution 800*480

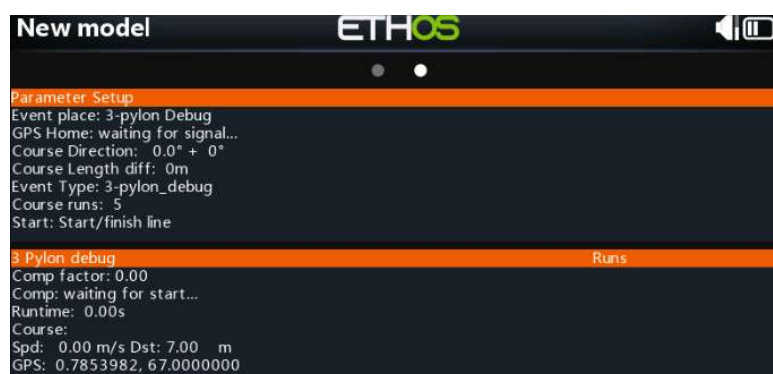
3. Known limitations

- Due to accuracy of GPS sensors (e.g. FrSky GPS ADV has horizontal accuracy approx 2.5m CEP, RCGPS-F3x 1.5 m CEP with SBAS) and telemetry latency the turn signals are not 100% precise, but still give a good F3X pylon-racing training experience. Moreover due to extreme acceleration in curves, which can achieve much above 4G assured for reliable GPS sensor operation, sensors can experience temporary satellite signal lock loss
- Sometimes there is a GPS-drift of given start point. In this case the whole course might drift to left/right or up/down some meters, because the turn positions are calculated in relation to the start point. This is especially important for start initiated by crossing over the start/finish line
- Max 14 fully defined and one "live" event sites are supported
- Application texts and menus are only in English. Speech announcements are given in language configured in the transmitter
- Application is resource demanding. There should not be many other widgets/system tools/tasks/sources running on transmitter otherwise accuracy of application can be compromised. It is valid also vice versa, so other applications can be affected by the GPS F3X Tracker
- For the sake of program speed, there is no check for 3 pylon courses if a plane is flying from pylon 1 to pylon 2 and not from pylon 1 to pylon 3, because it is easily detectable mistake

4. Installation

Unzip the installation package gpstrackp.X.x.zip, downloaded from the repository ([Releases gpsF3XTracker-Pylon-for-Ethos](#)), and place all files into directory /SCRIPTS on your transmitter. Folders gpstracap (keeping setup part) and gpstrackp (keeping operation part) should not be changed. Please note all modules, excluding locations.lua, are in the compiled form (*.luac) to be run as fast as possible.

Start the transmitter and configure two widgets – "GPS Pylon Tracker Setup" and "GPS Pylon Tracker" when a target model is selected. The application is capable to partly modify size of texts to size of widget windows, however, for accommodation of all information properly it is recommended to use at least half height & full wide layout for both setup and main widget (in such case widget titles should be switched off), or full height & half wide layout:



Note: upgrade from a previous program version can be done simply by replacing of all program modules by new ones. To mitigate potential run errors, it is strongly suggested to delete both widgets first and recreate them after replacement, checking and setting back all configuration items.

5. Configuration

- a) GPS sensor: set data rate to 10Hz = 0.1s (or higher if possible without lost of accuracy). Crosscheck names of available GPS sensors and rename, if needed. The application expects sensors coordinates, speed and satellites, if supported, with any of following name:
- Coord_names = {"GPS", "gps"}
 - Speed_names = {"GPS Speed", "GPS speed", "gps speed", "GSpd", "gspd"}
 - Sats_names = {"GPS Sats", "GPS sats", "gps sats", "GSats", "gsats", "GPS Satellites"}

For individual telemetry units:

1. RCGPS-F3x: use firmware 0.0.3d or newer. Create a DIY sensor "GPS Sats" with Application ID is 0x5111. Sensors coordinates, speed and satellites are supported
2. SM-Modellbau GPS-Logger 3: be careful – due to error in the Logger firmware v1.31 will Ethos wrongly recognize a sensor with Application ID 0x0860 and with name "GPS Satellites". Delete such sensor and create a new DIY sensor with Application ID 0x0870! It is probably corrected in the firmware v1.32, published on 23.01.2026. Sensors coordinates, speed and satellites are supported
3. FrSky GPS ADV: Sensors coordinates and speed are supported
4. Other GPS: Sensors coordinates and speed are supported

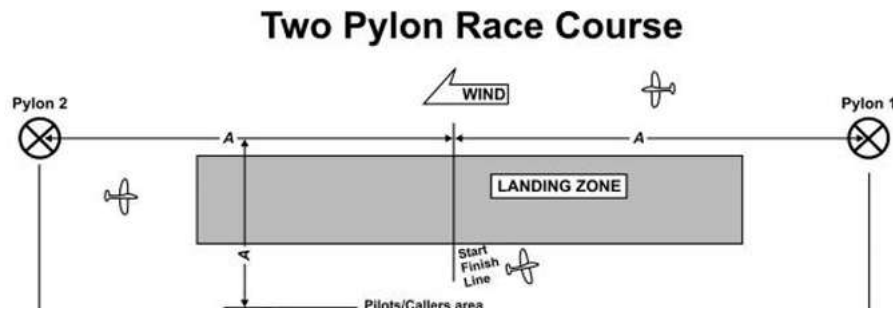
Note: Speed sensors must be configured in Ethos to give m/s!

Note: if the GPS sensor was bind (discovered) in the Ethos system version 1.6.1 or earlier, it is strongly recommended to delete it and discover again. It should fix various issues affecting Ethos before version 1.6.2

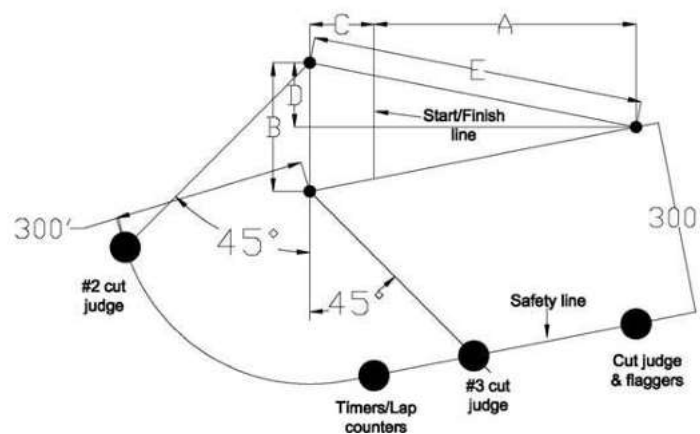
Note: not needed other telemetry values should be disabled in Ethos to speed up the telemetry transfer to transmitter. Also if possible, other telemetry sensors should not be used. A receiver polls 28 potential physical sensor units periodically, in an approximately 12 ms cycles (Physical IDs 0 - 27). One sensor unit can have multiple sensors types (Application IDs). Polling is dynamic, where active sensors are polled more often, without waiting for all the inactive ones, improving the data refresh rate. This means, if there is one active sensor it will be polled every 24ms, when there are two active sensors they will be polled every 36ms, etc. Lot of active sensors means problem for delivery of GPS coordinates as polling interval for individual sensor (= individual Physical and Application IDs) can be hundreds of milliseconds

- b) “GPS Pylon Tracker Setup” widget configuration:
1. Event place: any item from list of places in locations.lua file
 2. Course direction:

- a) 2-pylon: course bearing from the pylon 2 to the pylon 1 in degrees. Take direction by a compass from the pylon 2 to the pylon 1, e.g. 90° if the course direction is exactly East → West*



- b) 3-pylon: course bearing from center of line between pylons 2 and 3 and pylon 1



Standard distances: B=40m, C=30m, D=20m, E=180m

3. Course length difference: change of standard 180m course length <-100, +100>*
4. Course runs: number of course laps
5. Competition type: any type from supported types (2-pylon_training, 3-pylon_training, 2-pylon_debug, 2-pylon_debug)*
6. Start type: type of event timer initiation, Start race switch or Start/finish line**
7. GPS sensor: any item from list of supported units
8. Lock GPS Home position switch: any 2-position switch, physical or logical, or any other source with range <-100, +100>, mandatory***
9. Course direction difference management: source for real-time change of course direction, not mandatory, see chapter 11

* These items are available only for “Live Position & Direction event”, otherwise are locked as they are determined by event information from locations.lua file

** If the Start type configuration switch = “Start race switch”, the event timer starts immediately after the Start race switch is activated. If the Start type configuration switch = “Start/finish line” a new event starts by activation of the Start race switch, but the Tracker starts the event timer only when the plane goes first outside of the course (behind pylon 2 for 2-pylon course, behind both pylon 2 and 3 for 3-pylon course) and then it crosses the start/finish line from left to right.

*** Switch has to keep “high” value for permanent locking the GPS Home position, a momentary switch is not suitable. If a functional switch should be used, it must be configured as status switch

Event place	Live Position & Direction ▼
Course direction	0.5°
Course length difference	0m
Course runs	0
Competition type	2-pylon_training ▼
Edit event place	
Start type	Start race switch ▼
GPS sensor	RCGPS-F3x ▼
Lock GPS Home position switch	☒ FS1 ▼
Course direction diff mgmt	☐ Trim Throttle ▼

c) "GPS Pylon Tracker" widget configuration:

1. Start race switch: any 2-position switch, physical or logical, or any other source with range <-100, +100>, mandatory*
2. Logging: controls logging of event information
3. Flight correction factor management: Source for setting of the Flight correction factor during flight, not mandatory, see chapter 10
4. Flight correction factor: defines value for correction of flight position, 0 = no correction
5. Input debug GPS latitude and longitude: used for emulation of GPS input in debug mode (suggested analog sources elevator and rudder), not mandatory

* Switch can be state or momentary

Help	
Start race switch	☒ FS2 ▼
Logging	OFF ☐ ON ☒
Flight correction factor management	(x) Flighcomp ▼
Flight correction factor	0.20
Input debug GPS latitude	☒ Elevator ▼
Input debug GPS longitude	☒ Rudder ▼

6. Locations.lua

File Locations.lua, located in "/scripts/gpstrack/gpstrack" folder, keeps event location items in the format "Name of event site, home latitude, home longitude, course direction, course length difference, event runs, event type". Event types are:

- type 1: 2-pylon training
- type 2: 3-pylon training

-- type 3: 2-pylon debug
-- type 4: 3-pylon debug

Default Location.lua looks like below. You can edit it as per your needs, but please do not delete the first row, which defines “live” event location. Default event type in such case is 2-pylon training (type 1), however you can change it during configuration. Home position for “live” event location is defined from current GPS information. Please do not remove the last entry either (create new sites before that item) and keep its name "Last Entry". Max 15 event sites is supported.

```
{name = "Live Position & Direction", lat = 0.0, lon = 0.0, dir = 0.0, dif = 0, runs = 10, comp = 1},  
{name = "2-pylon Debug", lat = 0.0000000, lon = 0.0000000, dir = 0.0, dif = 0, runs = 5, comp = 3},  
{name = "3-pylon Debug", lat = 0.0, lon = 0.0, dir = 0.0, dif = 0, runs = 5, comp = 4},  
{name = "Site1 2-pylon training", lat = 31.2119998, lon = 121.4000015, dir = 100.8, dif = -41, runs = 9, comp = 1},  
{name = "Site2 3-pylon training", lat = 47.7019729, lon = 8.3558502, dir = 153.0, dif = 0, runs = 10, comp = 2},  
{name = "Last Entry", lat = 0.0, lon = 0.0, dir = 0.0, dif = 0, runs = 10, comp = 1}
```

Notes:

- home latitude and longitude is for 2-pylon events center of the course = the start/finish line
- home latitude and longitude is for 3-pylon events position of center of the start/finish line. It is crucial to set it exactly at the center of the line!
- course length of competition event types is defined as per F3X rules – 180m. You can change this default course length for a particular event site via item “dif” if needed – course is longer when value is positive and course is shorter when value is negative. It is not possible to change distance between pylons 2 and 3 for 3-pylon events. Debug items have their course length set internally to 80m
- *purpose of the "Live Position & Direction" event place is generally to test&try a new site and tune course parameters. If the site looks suitable and you expect to fly there more, it is necessary to create for it a new record in the Locations.lua (via Edit function) - all course parameters are then recorded. Otherwise the Tracker takes course direction, course length difference and competition type from the “Live Position & Direction” record in the Locations.lua, where by default is 'dir = 0.0, dif = 0, comp = 1' !*

You can edit the Location.lua file on a PC or edit individual records via embedded editor. Use button “Edit event place” on the site configuration screen to enter the editor. The original screen will be replaced by a new screen allowing editing of all parameters:

Event place	Site1 2-pylon training 
Latitude	31.2119998°
Longitude	121.4000015°
Course direction	100.8°
Course length difference	-41m
Course runs	9
Competition type	2-pylon_training ▼
Save	

Button “Save” saves modified site as below:

1. Creation of a new site from the “Live Position & Direction” event place – if list of sites is not full, new line with provided information will be created in the Locations table, just

- before the "Last Entry" line. Do not forget to change name of new event place. If list of sites is full, save operation will be refused and indicated by message "List of sites is full!"
2. Editing of an existing site – changed parameter(s) will be written into relevant site line in the Locations table

Editor can be closed by standard “back” widget Ethos button

Note: deletion of site lines in the Locations table and edition of parameters for “Live Position & Direction” event place is possible only via PC

7. Event modes

The application supports 2-pylon_training, 3-pylon_training, 2-pylon_debug event types. They behave differently as below:

- a) 2-pylon_training: it follows F3D rules, however only with 2 pylons. The run timer is started by a pilot via the Start race switch or by crossing the start/finish line, depending on the Start type switch. Time is then measured for individual concluded laps (visible only on display) and for whole training event (announced and visible on display when event is concluded)
- b) 3-pylon_training: it follows F3D rules fully. Time is measured and announced as above
- c) 2-pylon_debug: similar to 2-pylon_training, however it uses emulation of GPS input (via configured sources “Input debug GPS latitude and longitude”) for simulation of flight
- d) 3-pylon_debug: as above, but for course with 3 pylons

The actual status is indicated by individual rows in the "GPS Pylon Tracker" widget screen:

- a) Comp factor: defines value for correction of flight position, 0 = no correction
- b) Comp: “waiting for start...”, "started..." - just after switching the “Start race switch” on, "canceled..." - cancellation can be done by switching off/on/off of the “Start race switch”; information about concluded laps
- c) Runtime: time used for individual event
- d) Course: "center", "leftOutside", "leftInside", "rightOutside", "rightInside" – distance from the center is provided
- e) Spd, Dst: speed, distance from the center
- f) GPS: actual GPS position
- g) Runs: list of last events of the same type with their runtime

F3D: 2 Pylon debug	Runs
Comp factor: 0.00	1: 20:21
Comp: waiting for start...	
Runtime: 0.00s	
Course: rightOutside: 15.0	
Spd: 10.00 m/s Dst: 41.00 m	
GPS: 0.7853982, 101.0000000	

Announcements and sounds:

- Beep after switching the “Start race switch” on (600 Hz)
- Beep when crossing start/finish line from left to right (800 Hz) – only for events configured with Start type = Start/finish line
- Beep when crossing pylons plane(s) from inside to outside (1000 Hz)
- Beep when crossing the Start/Finish line from left to right in the last lap (1400 Hz)
- Lap number announcements
- Overall runtime at event end

8. Usage on a site

- a) Switch on RC system and give the GPS sensor enough time for full initiation and satellite detection – it can take 60+ seconds! For example for GPS-Logger 3 please wait till the orange LED is off and the green LED glows permanently or flashes. Open “GPS Pylon Tracker Setup” widget and select “Live Position & Direction” event place or any pre-configured place.
- b) For “Live Position & Direction event place”:
 1. Set “Competition type” configuration items if needed
 2. Set “Start type” configuration item if needed
 3. Go with your model to the home of the course (for 2-pylon events it is center of the course = start/finish line, for 3-pylon events it is center of start/finish line)
 4. Wait for stable information in the “GPS” row in the “GPS Pylon Tracker Setup” widget screen
 5. Take direction by a compass (2-pylon: course direction from the pylon 2 to the pylon 1 in degrees, 3-pylon: course direction from center of line between pylons 2 and 3 and pylon 1) and set it to the “Course direction” item (e.g. 90° if the course direction is exactly East → West)
 6. Lock the position with the “Lock GPS Home position switch” – such status will be indicated by change of item name from “GPS Home” to “GPS Home lck”
 7. Now your flight configuration is ready, you can go to start
- c) For other event places (pre-configured in the Location.lua file):
 1. Set “Start type” configuration item if needed
 2. Your other flight configuration is ready – all parameters are set in the Location.lua file
- d) Go to the "GPS Pylon Tracker" widget screen
The initial status is indicated by statement “waiting for start...”



F3D: 2 Pylon debug Runs
Comp factor: 0.00
Comp: waiting for start...
Runtime: 0.00s
Course: leftInside: -5.0
Spd: 10.00 m/s Dst: -5.00 m
GPS: 0.7853982, 55.0000000

- e) Start new event with the “Start race switch”, the timer will start depending on the “Start type” configuration item. If the Start type configuration switch = “Start race switch”, the event timer starts immediately after the Start race switch is activated. If the Start type configuration switch = “Start/finish line” a new event starts by activation of the Start race switch, but the Tracker starts the event timer only when the plane goes first outside of the course (behind pylon 2 for 2-pylon course, behind both pylon 2 and 3 for 3-pylon course) and then it crosses the start/finish line from left to right.

9. Logging

Logging of flight event information is supported. It should be used only if really necessary as it can affect performance of the application. Logging is disabled by default and has to be enabled in the "GPS Pylon Tracker" widget configuration. Flight events are written into log files “YYYYMMDD-Log” located in the folder /scripts/gpstrack/gpstrack.

Recording begins when a flight is started by the “Start race switch” and ends when an event is concluded. An initial log row looks e.g. like:

Start 16:06 Course direction:12.5°, Course difference:0m, luaRamAvailable:1758840B, Correction factor:0.0

Next rows have format as below, where:

- comp.state: provides information about current flight stage:
 - 2-pylon: 15 – waiting for plane crossing start/finish line from left to right, 20 – start competition timer, 25 – waiting for plane crossing right base from inside, 27 – waiting for plane crossing left base from inside, 28 – waiting for plane crossing start/finish line from left to right, 30 – end of event
 - 3-pylon: 15 – waiting for plane crossing start/finish line from left to right, 20 – start competition timer, 25 – waiting for plane crossing right base from inside, 26 - waiting for plane crossing left&up base from inside, 27 – waiting for plane crossing left&down base from up, 28 – waiting for plane crossing start/finish line from left to right, 30 – end of event
- GPS: gives current position of the plane
- Dist2home: gives current distance between the plane and home point (center of the course) in meters (negative value means the plane is on the left from the home point, positive value means the plane is on the right)
- Dir2home: gives current angle between the plane and home point in rads
- Course distance: gives current distance between the plane and home point in meters, projected into the event geometry. It gives information about position of the plane in relation to pylons
- Speed: speed of flight in m/s
- Sats: number of visible and used GPS satellites (available only for sensors providing that information)

comp.state, GPS: latitude, longitude, Dist2home, Dir2home, Course distance, Speed, Sats

10. Flight position correction

Due to latency of GPS sensor, telemetry latency and processing latency the turn signals are not 100% precise. The application has implemented basic mechanism aiming to compensate this issue. The function is controlled via value of item “Flight correction factor” in the "GPS Pylon Tracker" widget configuration:

- 0: correction is disabled
- 0.01 – 0.50: correction is enabled

Corrected position depends on a current flight speed and it is calculated as per formula below:
$$\text{CorrectedDistance} = \text{ReportedDistance} + \text{FlightcorrectionFactor} * \text{groundspeed}$$

The best value of the Flight correction factor must be found as it depends on hardware and software conditions. For example the Flight correction factor = 0.1, compensates the overall position inaccuracy related to one delayed report from a GPS sensor set to 10 Hz, so ideally reporting each 0.1s. In a case of speed 100 km/h, compensation is 2.8m, with speed 150 km/s it is 4.2m. The Flight correction factor = 0.5 compensates 5 delayed reports and compensation is 14m respectively 21m for the same speeds.

You can set the Flight correction factor manually or you can set its best value during test flights:

- There is new configuration item of type Source "Flight correction factor management". If it isn't configured, it is possible to change the configuration item "Flight correction factor" manually, otherwise the item "Flight correction factor" is managed by the configured source
- Value of the item "Flight correction factor management" must be in range 0-50% to achieve range 0-0.5. I advise to create a Var, e.g.:

Value	20.0%		
Range	0.0%	-	50.0%
Values	Slider right ▼	-20.0%	
+ Add a new value			
Actions			
Always on ▼	Add(+) ▼	100.0	
Always on ▼	Divide(/) ▼	4.0	

- Change of value is in the Tracker accepted only during status "waiting for start...". However value of the source can be changed anytime and the Tracker will accept it when goes again into the status "waiting for start...".
- Change of value is announced by voice, be patient as Ethos chains voice messages
- Last value is stored in the item "Flight correction factor", you can see it in the operation widget
- After landing configure the item "Flight correction factor management" as empty – in this way you will fix the found best value of the correction factor. It is expected to use this function only when a new model is configured or its hardware (sensors, receiver) has changed, so the "Flight correction factor management" configuration item is not mandatory and should stay generally empty to avoid unexpected change of the correction factor!

11. Management of course direction difference

Accuracy of setting the course direction is crucial for correlation between actual position and the course frame (vertical planes at pylons). Any mistake in setting of the course direction means wrong indication in crossing of pylons. Depending on how far you fly from the axis of the course, inaccuracy in indication can be couple of meters even for relative small error in course direction (error 5° means 0.4m inaccuracy 5 meters from the axis, but 2.6m when you fly 30 meters from it). Wrong course direction can be identified by a permanent “move” of pylons to the left or right. When crossing of all pylons is reported to the right of physical pylons, try to decrease the course direction and vice versa.

Standard course direction for individual event can be permanently changed via configuration and reflected in the related event location item in the Locations.lua. For the “Live Position & Direction” event place there is possibility to change its course direction in the range <-10, +10> degrees during flight without landing.

For the feature to run you need to set a source for the “Course direction difference management” configuration item. It is suggested to use e.g. a free trim configured with Easy mode and Fine step – in such case each trim move will change the direction by one degree in positive or negative manner. A Var with range -20%-20% with any suitable source is also very good and universal solution.

Change of the direction is possible only when the position is locked with the “Lock GPS Home position switch” – change is confirmed by a voice announcement and indicated on the “Course Direction” row of the “GPS Pylon Tracker Setup” widget.

How to use this function: Fly near axis of the course, observe where the detection occurs, and then fly in the area far from the axis. Observe the difference and try to correct if necessary, until you feel you have found the correct course direction.

Notes:

- do not use this function without tuning of the Flight correction factor first! Otherwise you can face inaccurate position information!
- course direction made by this feature is not considered as permanent and it isn't recorded. If the change in the course direction should be set permanently, configure it after landing in the Locations.lua
- if the source configured for the “Course direction difference management” item isn't zero/neutral at a moment of locking with the “Lock GPS Home position switch”, its value is considered and reflected as course direction difference
- if the “Course direction difference management” source changes its value after start of a “Live Position & Direction” event, the event is canceled and it goes into "waiting for start..." status
- it is expected management of course direction will be used only rarely for previously unknown sites or for testing, so the “Course direction difference management” configuration item is not mandatory and should stay generally empty to avoid unexpected change of the course direction!

12. Change log

V1.1:

- only lap number without lap time is announced to mitigate delay in indication of reaching pylon 1

V1.2:

- support of standard 3 pylon courses
- enhancement by support of start sequences initiated by the Start race switch or by crossing the start/finish line

13. Development plan

It is probable there will be necessary to change or enhance some parts. Do not hesitate to comment and come with ideas, preferably via an Issue in the GitHub repository ([New issue gpsF3XTracker-Pylon-for-Ethos](#))

14. License

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